

Model-based Smoothing for Near-Monotone Functions: Two Possible Approaches

Ziang Zhang, Patrick Brown, Jamie Stafford

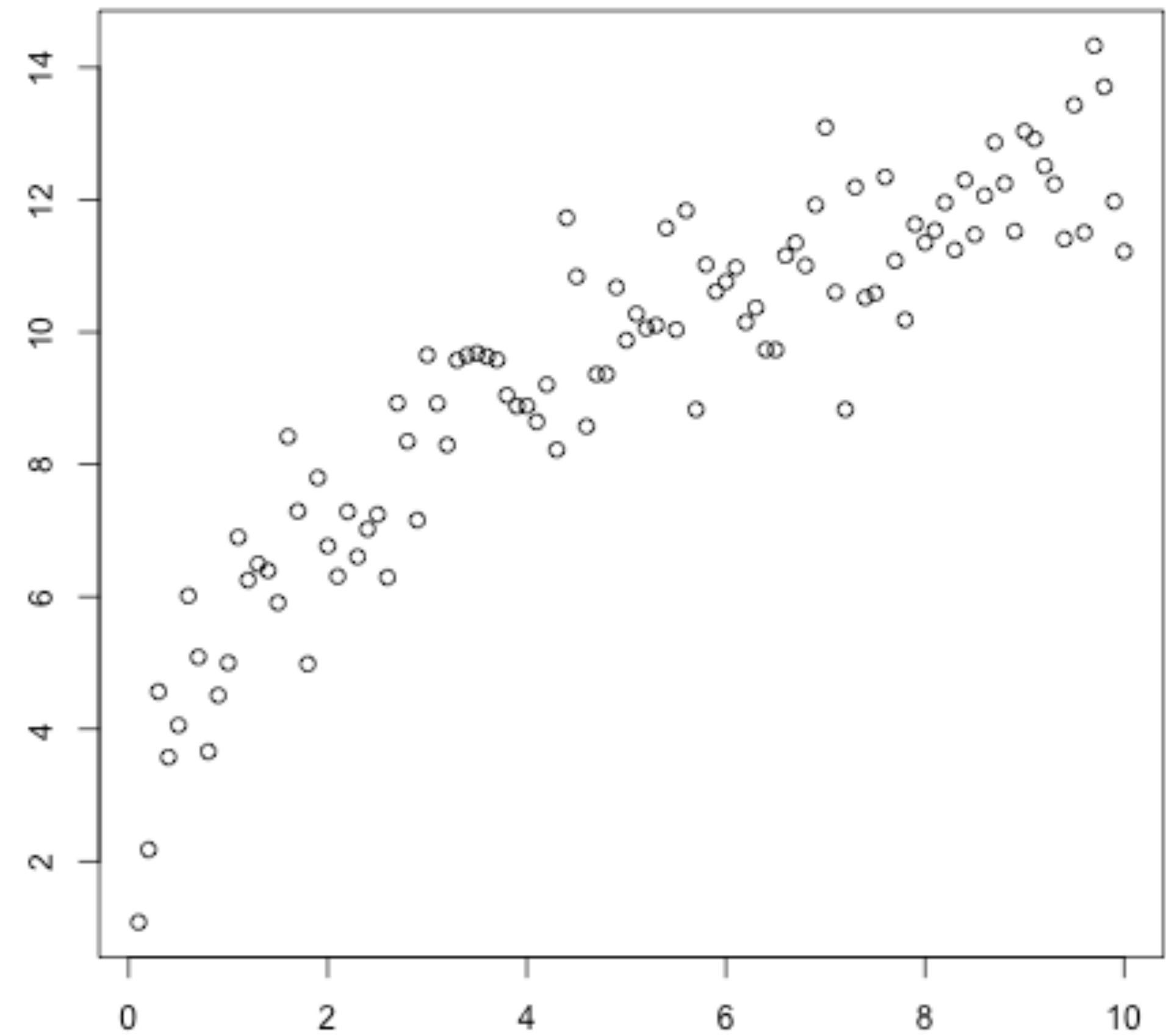
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The classical smoothing problem:

- $y_i = f(x_i) + \epsilon_i$
- f is the unknown function of interest
- f assumed **smooth**

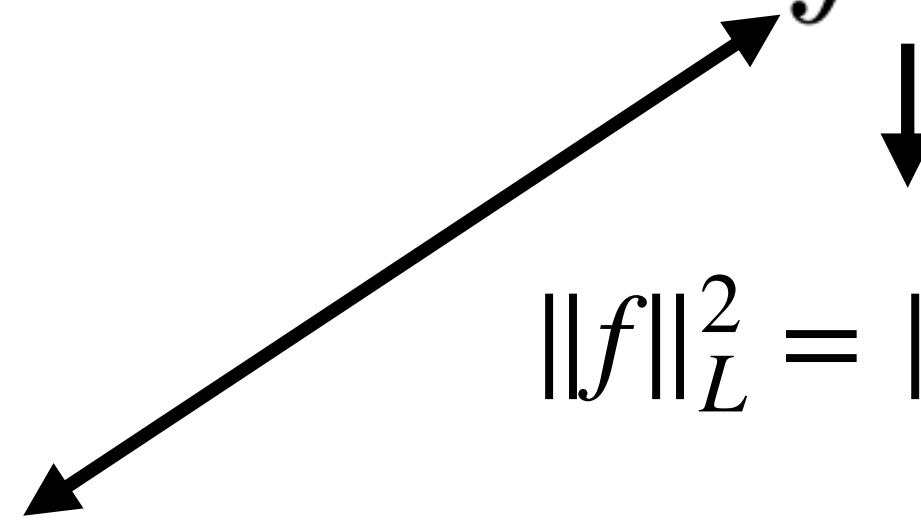
BUT, what does smooth mean here?

- Mathematicians say $f \in C^\infty$
- Statisticians look at certain **norm** $\|f\|_L$



How Statisticians View Smoothing

$$\sum_{i=1}^n (y_i - f(x_i))^2 + \lambda \int (f''(x))^2 dx.$$


$$\|f\|_L^2 = \|Lf\|_2^2 \quad \text{with} \quad L = D^2$$

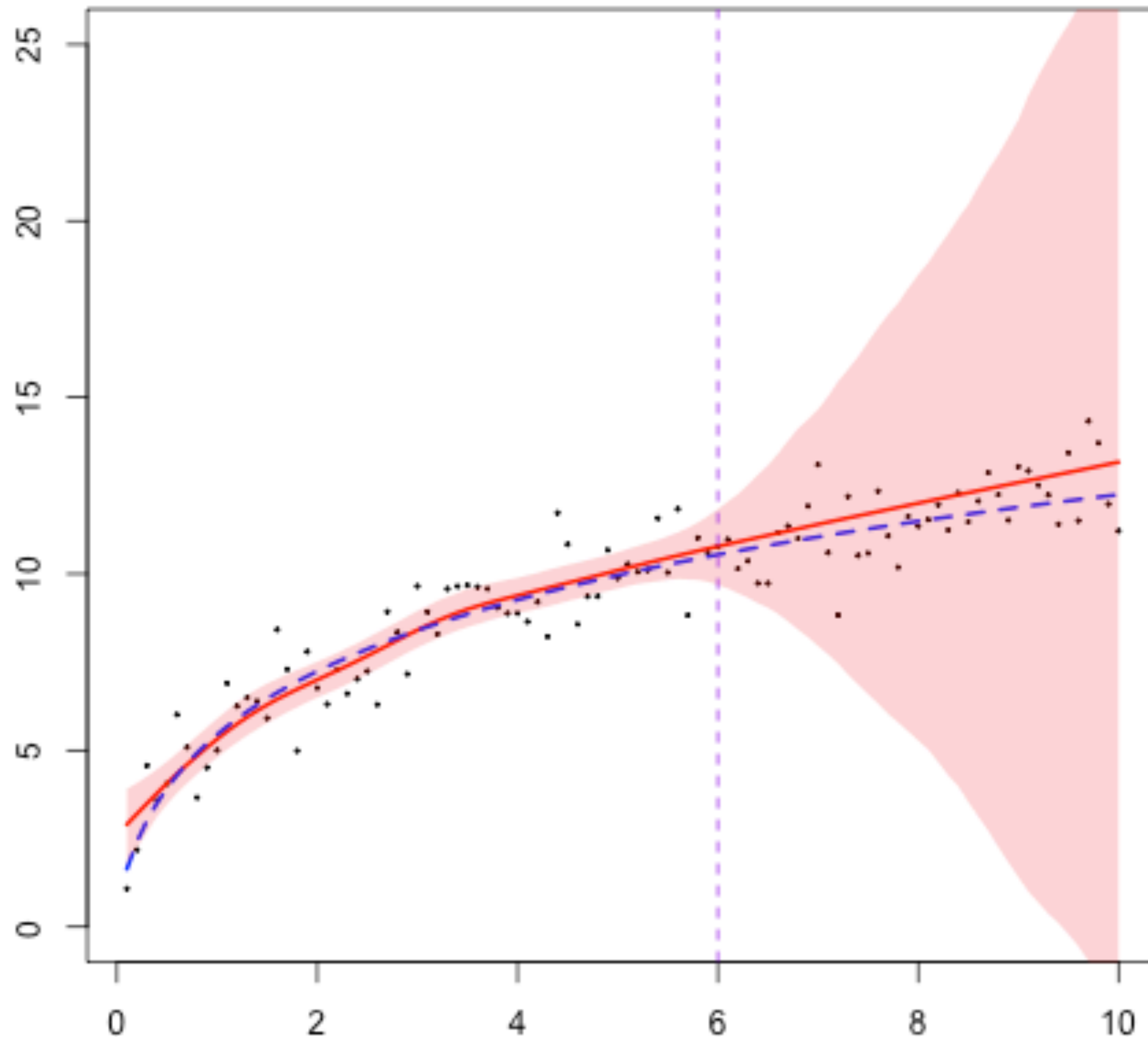
$$D^2 f(x) = \sigma \xi(x), \sigma = 1/\sqrt{\lambda} \quad \longleftrightarrow \quad \text{Integrated Wiener Process: } f \sim \text{IWP}_2(\sigma)$$

- $\text{Null}\{L\} = \text{span}\{1, x\}$
- Linear functions are not penalized
- Smooth means $f(x) \approx a + bx$

Model-based Smoothing

- Smoothing can be viewed as doing shrinkage
- f is shrunk toward **the base model** (a set of simple functions)
- E.g. $f \sim \text{IWP}_2(\sigma) \rightarrow$ base model **Null** $\{L\} = \text{span}\{1, x\}$
- Base model specifies the “**direction**” of the shrinkage
- What if we know $f \approx a + bm(x)$ for some $m(x)$?

Fitting an IWP2 to the data



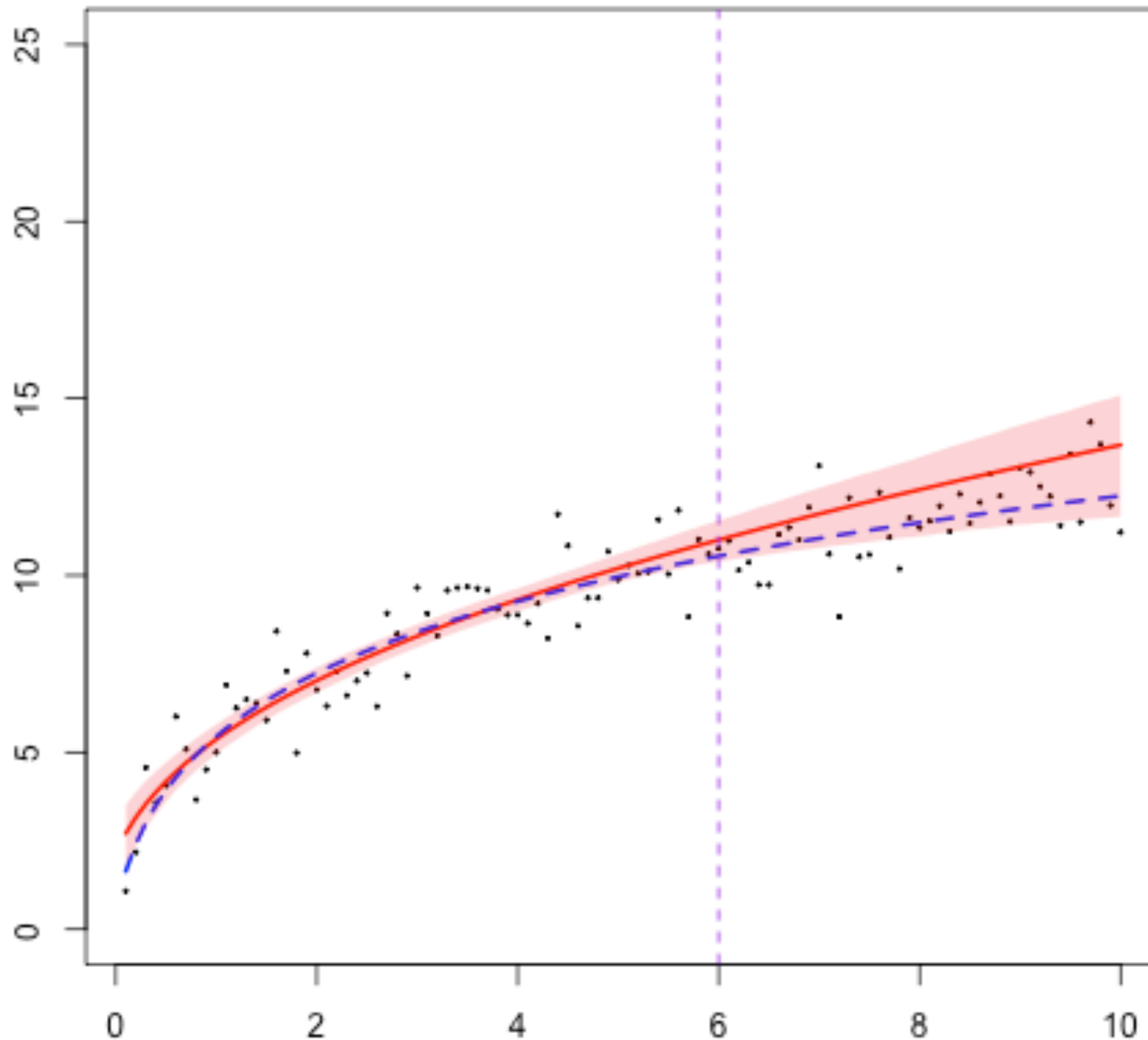
- f is highly nonlinear at $x \approx 0$
- $\text{IWP}_2(\sigma)$ estimates a large σ
- IWP gives a linear prediction
- With HUGE prediction interval

An Intuitive Approach:

Transforming the IWP2: t-IWP2

- Now define $x^* = m(x)$ $f(x) := \tilde{f}(m(x)) := \tilde{f}(x^*)$
- Assume $\tilde{f}(x^*) \sim \text{IWP}_2(\sigma)$
- The base model becomes $\text{span}\{1, x^*\} = \text{span}\{1, m(x)\}$
- We call it the transformed-IWP2, $f \sim t\text{IWP}_2(\sigma)$

Fitting an $t\text{IWP}_2$ to the data



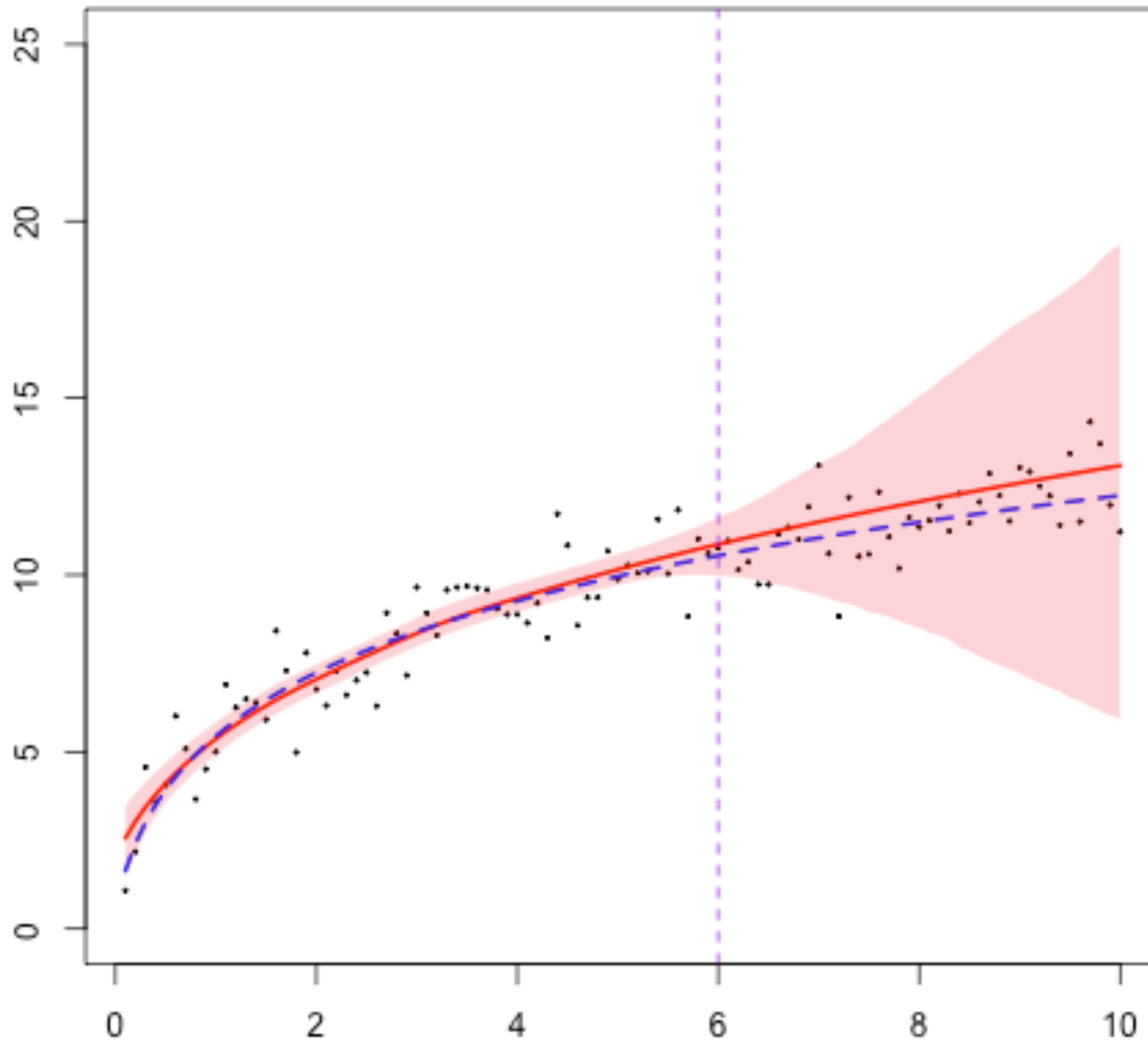
- Using the $t\text{IWP}_2(\sigma)$ prior for f
- The generating $m(x) = \log(x)$
- Prediction: $a + b \log(x)$

There is another approach...

Why don't we define a new operator?

- Recall if $Lf = \sigma\xi$, the base model of f is $\text{Null}\{L\}$
- Define $L = D^2 - \alpha(x)D$, where $\alpha(x)$ is Lebesgue square integrable.
- Then $\text{Null}\{L\} = \text{span}\{1, D^{-1} \exp(D^{-1}\alpha)\}$
- Let $m = D^{-1} \exp(D^{-1}\alpha)$, $\alpha = D^2m/Dm$ (relative curvature)
- Any (reasonable) monotone $m(x)$ can be written this way (*Ramsey, 1997*)
- We call it the monotone-induced GP, $f \sim \text{mGP}(\sigma)$

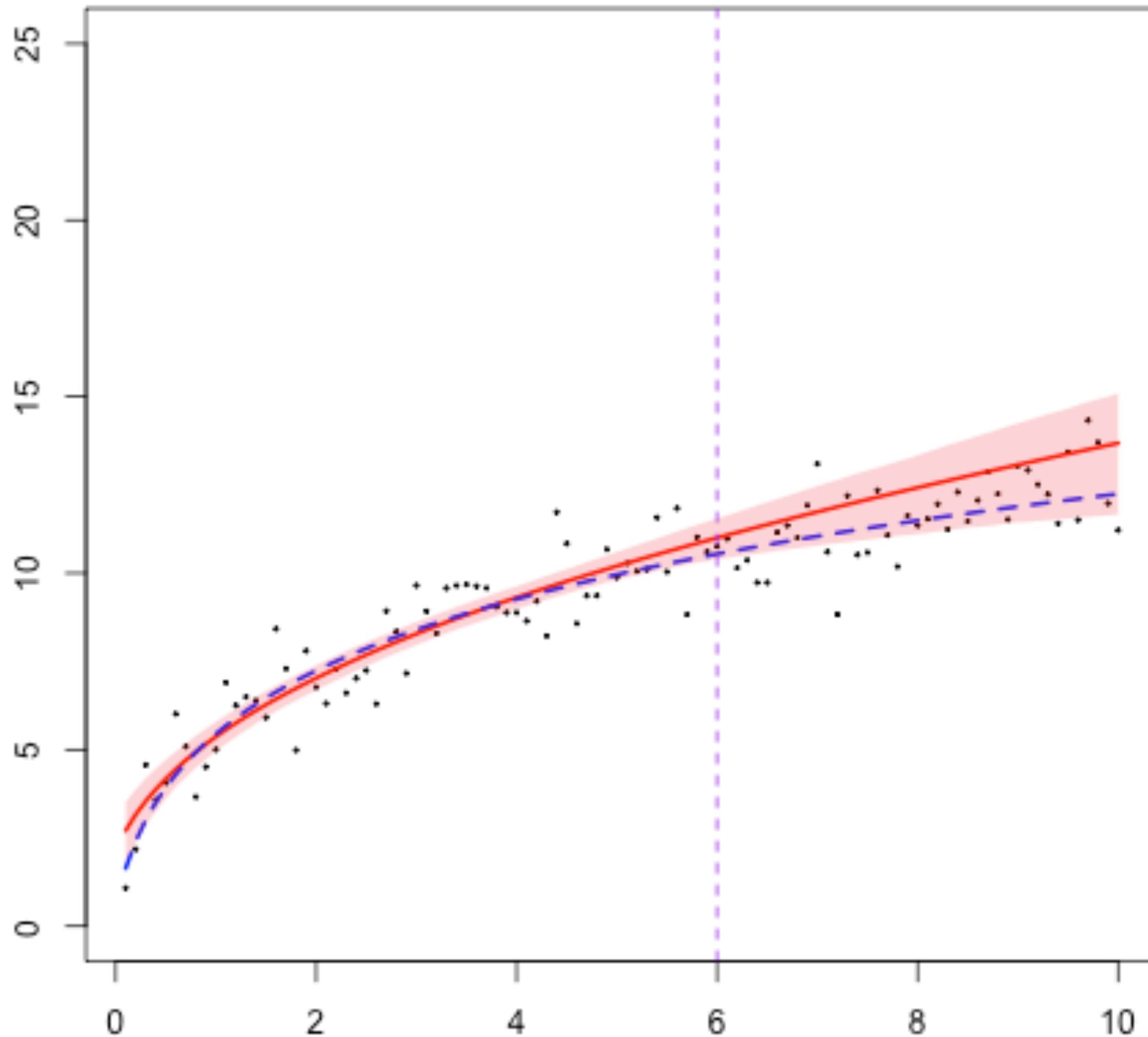
Fitting a mGP to the data



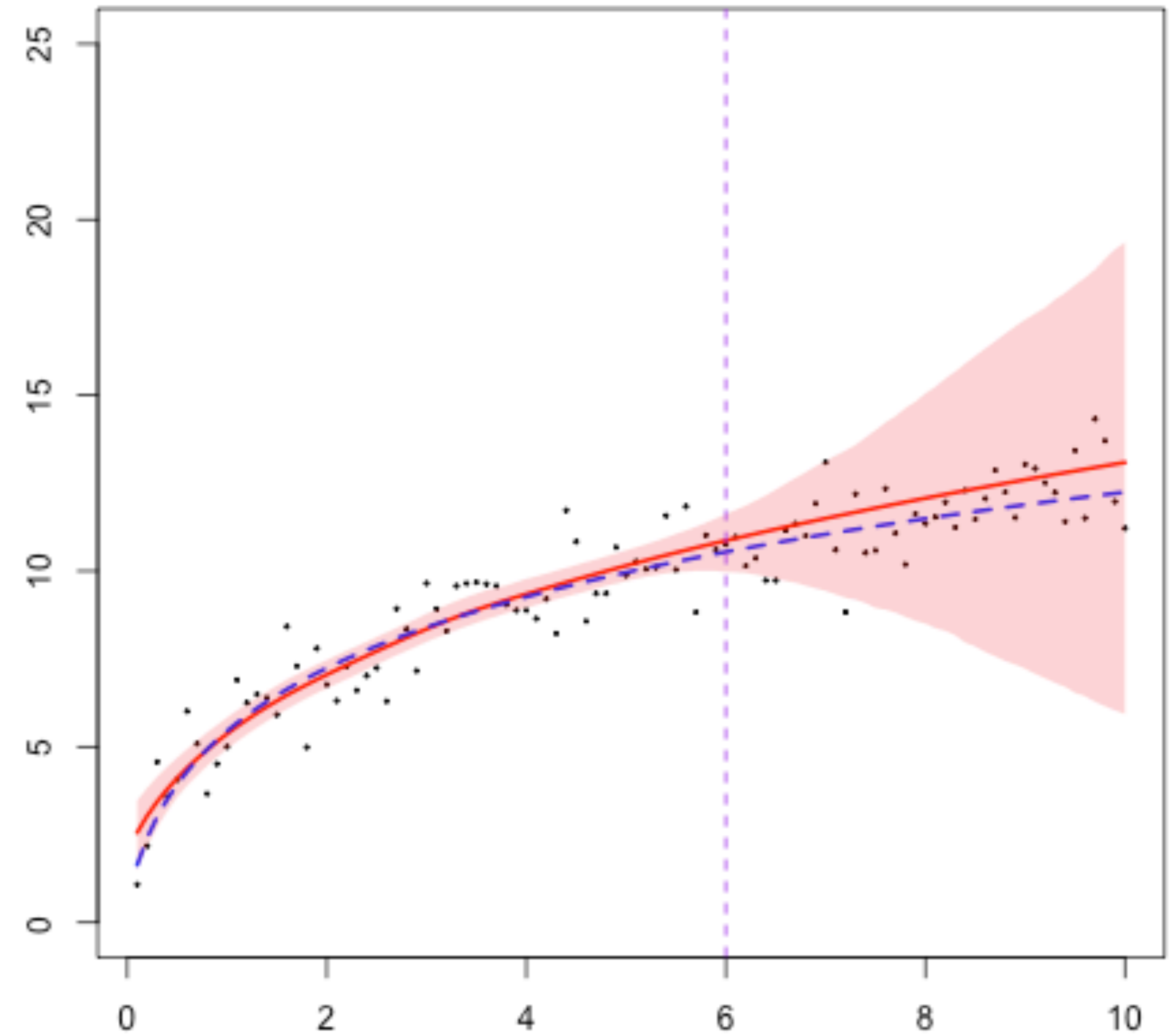
- Using the $\text{mGP}(\sigma)$ prior for f
- The generating $m(x) = \log(x)$
- Prediction: $a + b \log(x)$

Comparing the two new approaches:

t-IWP2



mGP



Comparing the two approaches..

- The tIWP and mGP provided similar posterior mean
- The prediction interval from mGP is wider
- Which approach should we use?
- Well, we can show that tIWP is **equivalent** to a particularly type of mGP

tIWP2 is a time varying mGP

Proposition 1 (t-IWP as time varying mGP): If $f \sim t\text{IWP}(\sigma)$ generated by a monotone function $m(x)$ and define $\alpha(x) = m''(x)/m'(x)$ then

$$Lf(x) = \sigma(x)\xi(x)$$

where $L = D^2 - \alpha(x)D$ and $\sigma(x) = \sigma[m'(x)^{3/2}]$.

Some implications...

Transforming coordinate imposes implicit assumption

- When $m(x) = \sqrt{x}$, $m'(x) = \frac{1}{2\sqrt{x}}$
- $\sigma(x) = [\sigma m'(x)^{3/2}] \approx 0$ when x is large
- $tIWP_2(\sigma)$ assumes there is **no deviation from base model** when x is large
- This is a **HUGE** assumption!

Looking at the PSD (Zhang et.al, 2024):

- σ is not interpretable across different models
- Prior elicitation based on the h-step predictive standard deviation (PSD)

$$\sigma_x(h) = \text{SD}[f(x+h) | f(u) : u \leq x].$$

- Unlike IWP_2 , this quantity **depends on both** x and h for $t\text{IWP}_2$ and mGP
- What will happen if $x \rightarrow \infty$?

Two different limits...

Corollary 1 (Limiting PSD of M-GP):

Assume $\alpha(x) = -\frac{1}{a(x+c)}$ where $a \neq 0$. For the M-GP defined by $\alpha(x)$

$$\sigma^2(h) = \lim_{x \rightarrow \infty} \sigma_x^2(h) = \frac{\sigma^2 h^3}{3}.$$

Corollary 2 (Limiting PSD of t-IWP2):

Assume $\alpha(x) = -\frac{1}{a(x+c)}$ where $a \geq 0$. For the t-IWP2 defined by a monotone transformation $m(x)$ such that $m'(x)/m''(x) = \alpha(x)$, the limiting PSD is

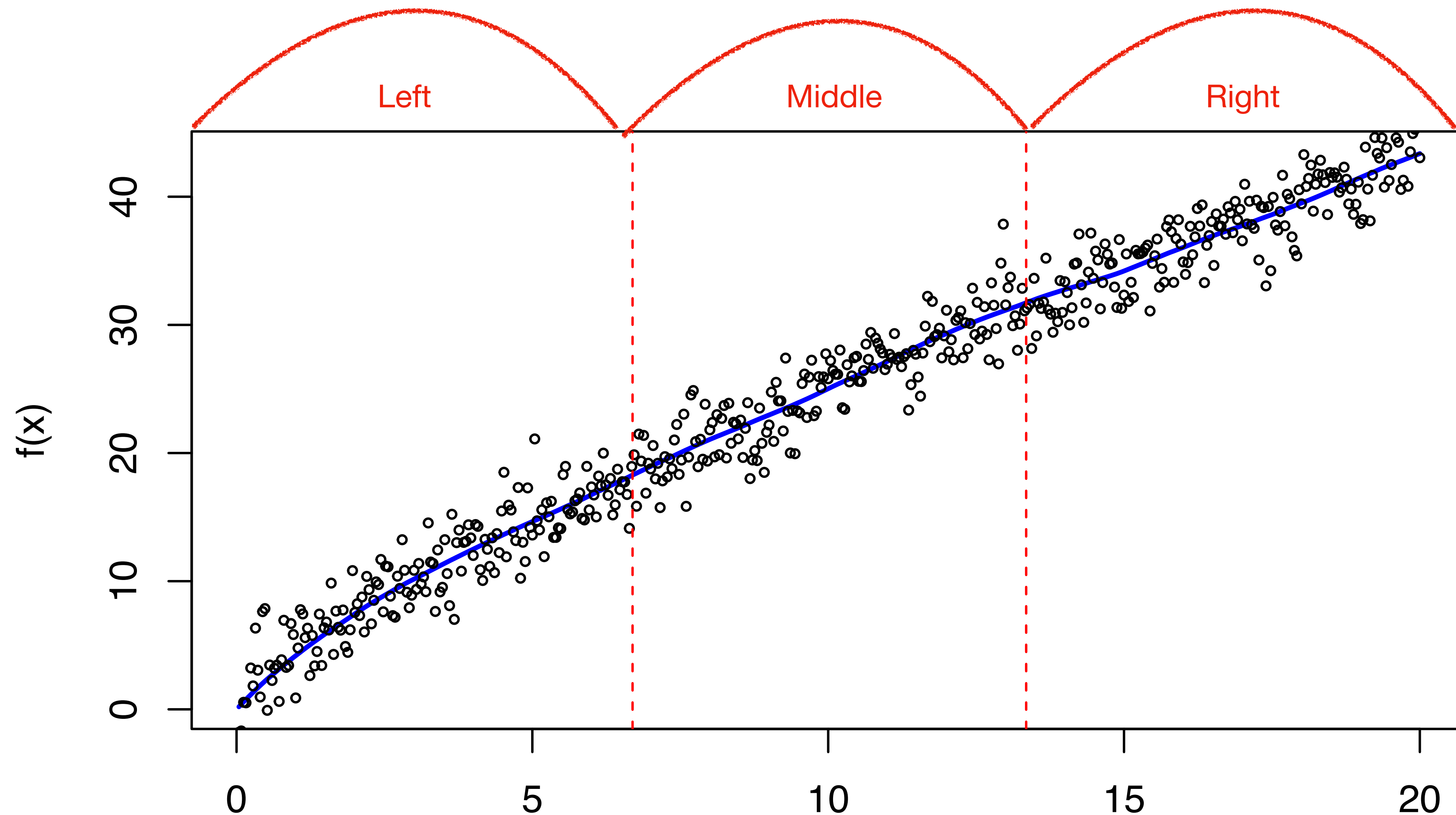
$$\sigma^2(h) = \lim_{x \rightarrow \infty} \sigma_x^2(h) = 0.$$

A Simulation Study

Which model is more robust to model-misspecification?

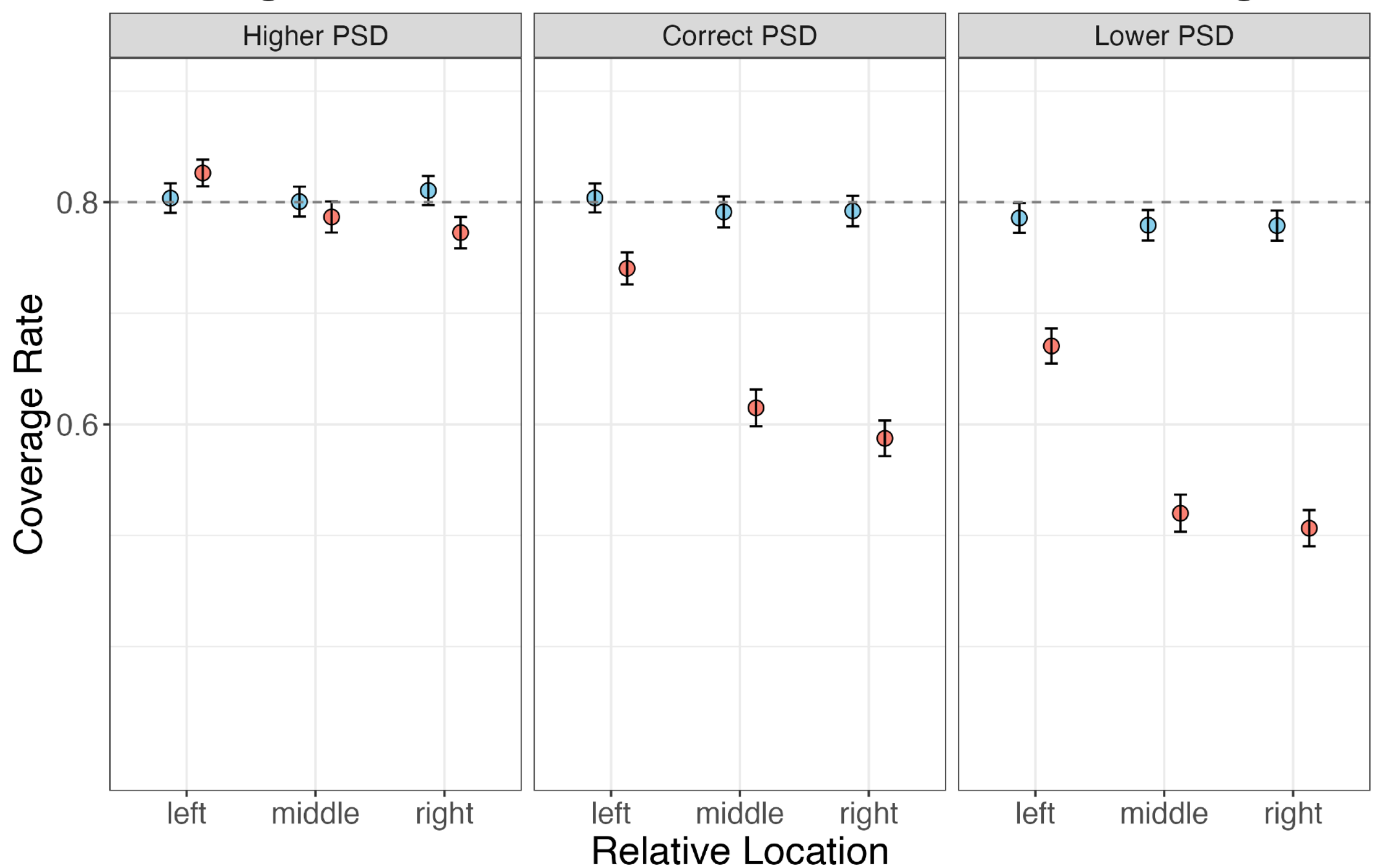
- In **scenario A**, we consider:
 - $f \sim \text{mGP}$ induced by $m(x) = \sqrt{x}$, with $\sigma_0(20) = 1$
- In **scenario B**, we consider:
 - $f \sim t\text{IWP}_2$ induced by $m(x) = \sqrt{x}$, with $\sigma_0(20) = 1$
- For both case, we compute the point-wise coverage rate and interval width.
- Exponential priors used on $\sigma_0(20)$ with **higher, correct** or **lower** medians.

A Stratified Comparison...

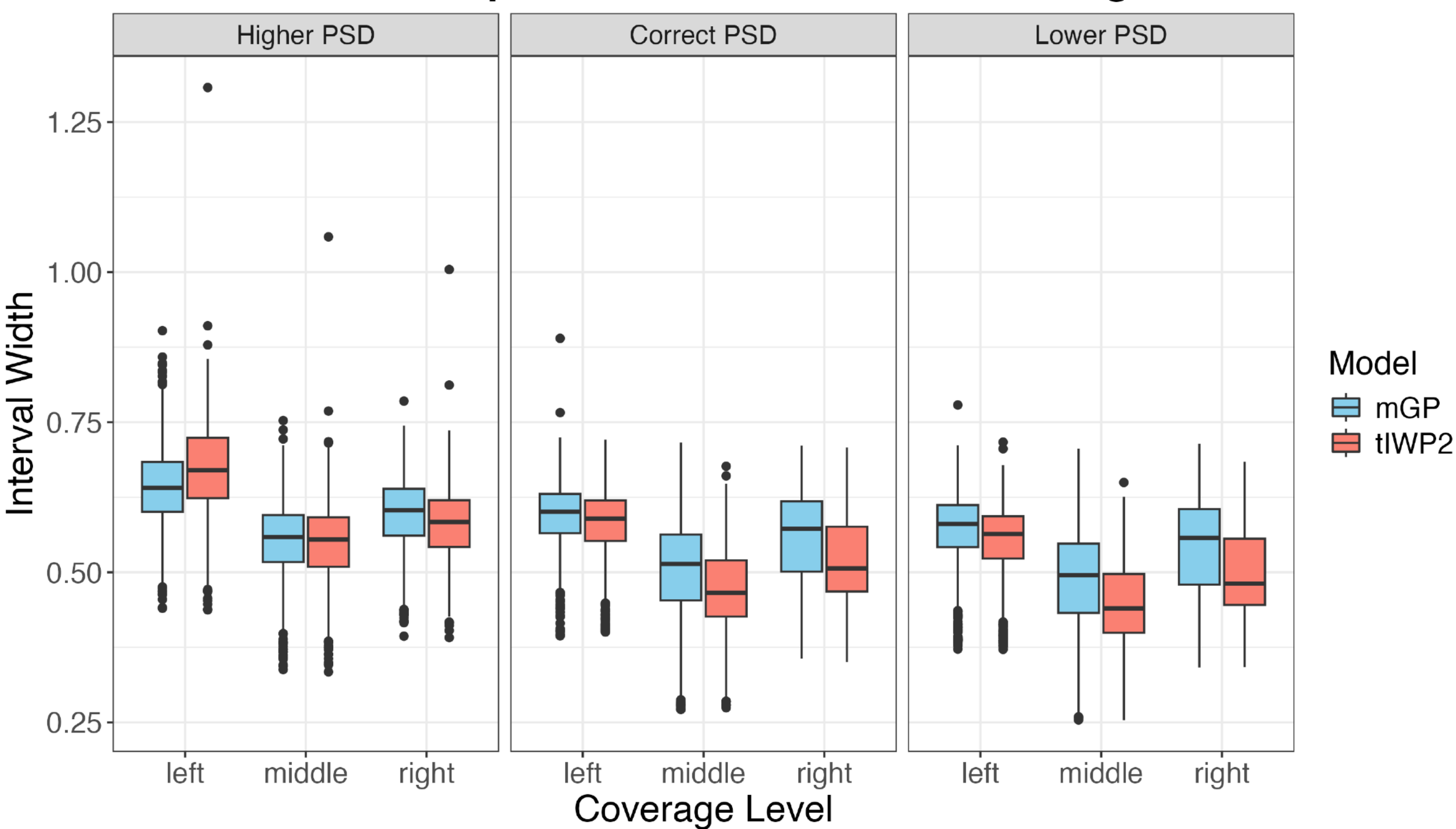


Scenario A

Coverage Rate with Error Bars for Different PSD Settings

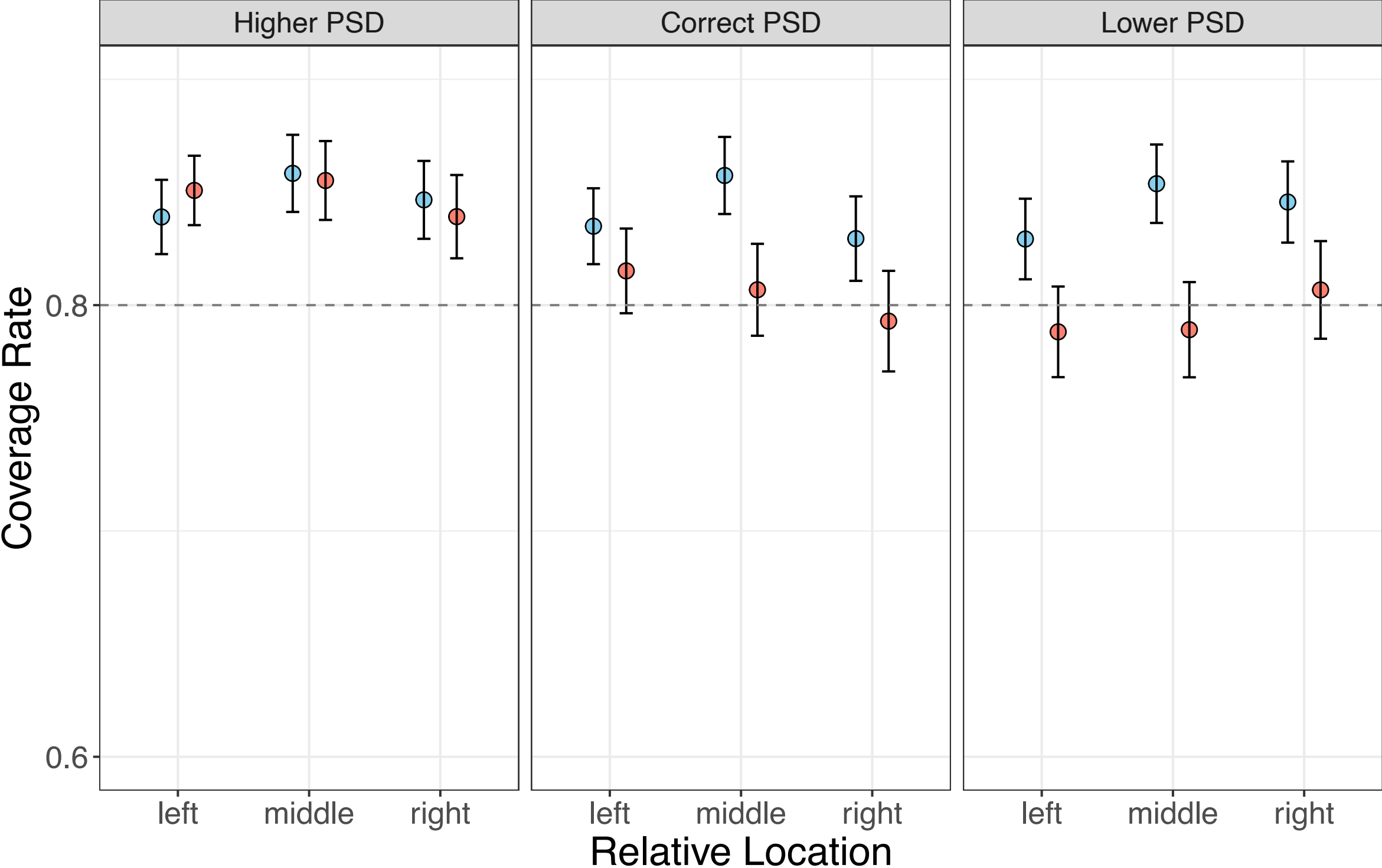


Interval Width Comparison for Different PSD Settings

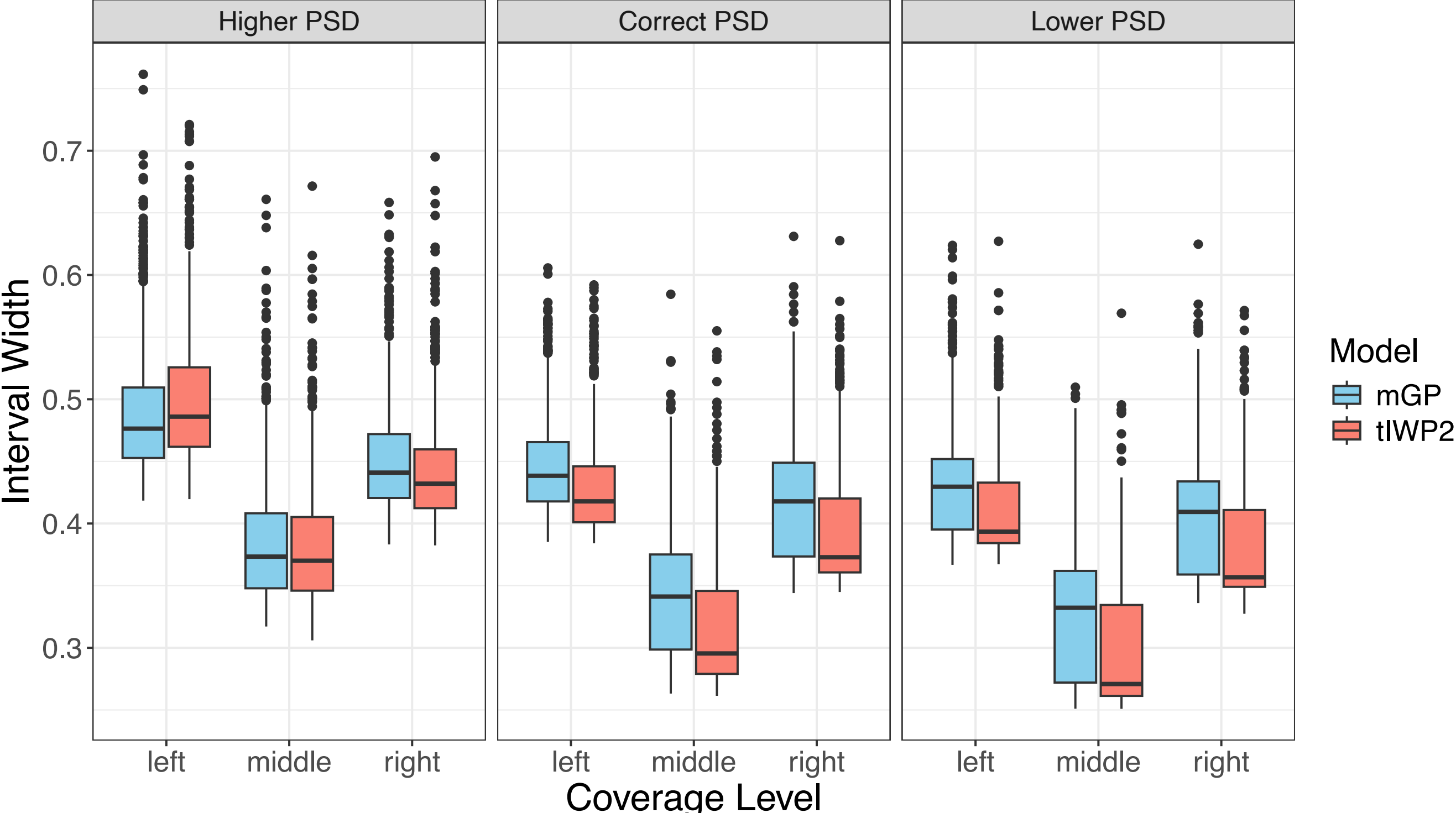


Scenario B

Coverage Rate with Error Bars for Different PSD Settings



Interval Width Comparison for Different PSD Settings



Conclusion

- Both approaches can be used for near-monotone smoothing
- Both are computationally efficient through state-space or FEM
- Which one is better? Depending on how does f deviate from the base model
- Under model-misspecification, mGP appears more robust

Reference

Ramsay, J. O. (1998). Estimating smooth monotone functions. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 60(2), 365-375.

Zhang, Z., Stringer, A., Brown, P., & Stafford, J. (2024). Model-based smoothing with integrated wiener processes and overlapping splines. *Journal of Computational and Graphical Statistics*, 1-13.

- For more details about this work, see:
 - aguerozz.github.io/summary_M_GP